

NAME : M NIKHIL SAI
 SRN : PES1UG24CS253
 SECTION : E

Req Id	Type	Description	Priority	Acceptance Criteria	Rationale	
Functional Requirement 1	Functional	The system will allow members to search the ISBN repo and place hold requests to enter FIFO	High	Pass: User is assigned the correct FIFO queue position. Fail: User exceeds the maximum allowed holds but is still assigned a queue position.	Core reservation functionality.	
FR 2	Functional	The system will display the current position of a student member in FIFO waitlist for a reserved book	High	Pass: System displays the student's correct current position in the waitlist. Fail: Displayed position does not match the student's actual FIFO position.	Allows members to track their reservation status.	
FR 3	Functional	The system will notify the next eligible student member when a held book becomes available	High	Pass: When a held book becomes available, the student at the front of the queue receives a notification. Fail: The notification is not sent to the eligible student.	Required for automated hold management.	
FR 4	Functional	The system shall allow the head librarian to update the availability status	High	Pass: A change made by the Head Librarian to a copy's availability is reflected in the	Librarians need to maintain accurate inventory status.	

				system. Fail: The previous availability status remains displayed after the update.		
FR 5	Functional	The system shall record overdue fine transactions associated with student members	Medium	Pass: The system records the member, loan/book reference, fine amount, and transaction status for an overdue fine. Fail: Any required fine-transaction field is missing or incorrect.	Required to track overdue fine transactions.	
Non Functional Requirement 1	Non Functional	The catalog search query shall return results for 50,000 titles in less than 150 ms	High	Pass: Catalog searches for 50,000 titles return paginated results in less than 150 ms under the specified test conditions. Fail: The response time is 150 ms or greater or the required security conditions are not satisfied.	Ensures responsive and secure catalog searching.	
NFR 2	Non Functional	The system shall preserve FIFO queue ordering when multiple valid hold requests are submitted for the same unavailable book	High	Pass: 100 sequential valid hold requests are assigned	Prevents incorrect reservation ordering and ensures fair allocation.	

				queue positions in the same order in which they were accepted. Fail: Any valid request receives a queue position inconsistent with its acceptance order.		
--	--	--	--	---	--	--